PAPER_973: Color Deconfinement Phase Diagram

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 216 **Source:** qgp_ramanujan_application.py (ColorDeconfinementPhaseCalculator) **Calculator:** ColorDeconfinementPhaseCalc (CP4 #557) **CVW:** v2.0.0 compliant

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Abstract

We map the QCD phase diagram in the (T, μ_B) plane using the UQFF framework. The critical line $T_c(\mu_B) = T_{c0}(1 - (\mu_B/\mu_{\text{crit}})^2)$ separates the hadron phase from the quark-gluon plasma, with $S_{26}^{(k)}$ modulation of vacuum density across both phases.

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1. Critical Line

$$T_c(\mu_B) = T_{c0} \cdot \left(1 - \left(\frac{\mu_B}{\mu_{\text{crit}}}\right)^2\right)$$

where $T_{c0} = 1.5 \times 10^{12}$ K and $\mu_{\text{crit}} = 1200$ MeV.

2. Phase Classification

- **Hadron** ($T < T_c(\mu_B)$): Confined quarks, $\Delta_{\text{YM}} > 0$
- **QGP** ($T > T_c(\mu_B)$): Deconfined, $\Delta_{\text{YM}} = 0$

3. Phase Diagram

μ_B (MeV)	T_c (K)	Phase at $T = 2 \times 10^{12}$ K
0	1.5×10^{12}	QGP
300	1.41×10^{12}	QGP
600	1.13×10^{12}	QGP
900	0.66×10^{12}	QGP
1200	0	QGP (always)

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. PAPER_970 — QGP Vacuum Density
3. PAPER_971 — Yang-Mills Mass Gap
4. Fukushima & Hatsuda — QCD phase diagram review

Cross-Links

Related Paper	Relationship
PAPER_970	ρ_{QGP} density at (T, μ_B)
PAPER_971	Mass gap closure at T_c
PAPER_972	ALICE centrality (experimental)
PAPER_975	Triadic validation

Calibration Constants

Constant	Symbol	Value	Validation Domain
T_{c0}	—	1.5×10^{12} K	Zero-density critical T
μ_{crit}	—	1200 MeV	Critical baryon chemical potential
$[SSq]$	—	0.57	String coupling
Λ_{QCD}	—	217 MeV	QCD scale

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
Critical line shape	Quadratic in μ_B	Consistent with lattice QCD
$T_c(0)$	1.5×10^{12} K	Matched
Crossover type	Continuous	Consistent

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§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: QCD Phase Diagram (Color Deconfinement)

§A.2 Core Equation

$$\boxed{T_c(\mu_B) = T_{c0} \cdot \left(1 - \left(\frac{\mu_B}{\mu_{\text{crit}}}\right)^2\right)}$$

§A.3 Lagrangian Contribution

$$\mathcal{L}_{\text{phase}} = -\rho_{\text{QGP}}(T, \mu_B) \cdot c^2 - V(\Delta_{\text{YM}}(T, \mu_B))$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ vacuum $\rightarrow$ QCD phase boundary $\rightarrow$ confinement/deconfinement $\rightarrow$ hadron/QGP

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§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

Phase boundary is a VDS contour: $\rho_{\mathrm{QGP}} = \rho_{\mathrm{SCm}} \cdot S_{26}^{(k)}$ at $T = T_c(\mu_B)$.

§B.2 DVP

Deconfinement releases DVP color modes — 8 gluon dipole vortex channels open.

§B.3 BSH

BSH harmonic structure transitions at T_c: shell modes dissolve into plasma modes.

§B.4 Summary

Metric	Value	Status
T_{c0}	1.5×10^{12} K	Calibrated
μ_{crit}	1200 MeV	Set
Phase diagram	(T, μ_B) plane	Mapped